Mistake 2:

> "Measurement doesn't affect system"

❌ Wrong
✔️ Measurement **collapses quantum state**

Mistake 3:

> "Bell states are random"

❌ Wrong
✔️ They are **perfectly correlated**

If You Want Next Step

Tell me:

1. Want ↳ **diagram explanation (visual intuition)**?
2. Want ↳ **minimal code version (exam-ready)**?
3. Want ↳ **math derivation (Dirac notation)**?
4. Want ↳ **debugging workflow (why teleportation fails)**?

I'll go deeper accordingly.

I didn't get please explain full again

You're not missing something small—you're missing the **mental model**. I'll rebuild everything from zero, slower, and map every line of code to a **single simple story**.

🔴 STEP 0 — Think like this (very important)

Forget physics.

Think:

- A **qubit = a coin**
- But unlike a normal coin, it can be:
  - Heads (0)
  - Tails (1)

- **Spinning (both at same time)**

🔴 STEP 1 — What is Bell State? (core idea)

👉 Two coins become **magically linked**

If one is:

- Heads → other is Heads
- Tails → other is Tails

Even if far apart.

This is called **entanglement**

🔴 PART 1 — CODE EXPLANATION (VERY SLOW)

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1. Imports

```python
from qiskit import QuantumCircuit
```

👉 Tool to create quantum program

```python
from qiskit_aer import AerSimulator
```

👉 Fake quantum computer (simulator)

```python
from qiskit.quantum_info import Statevector
```

👉 Shows exact internal quantum state

---

2. Simulator

```python
sim = AerSimulator
```

👉 Create virtual quantum machine

---

3. Function

```python
def create_bell_state state_type
```

👉 Function → creates different Bell states

```python
qc = QuantumCircuit 2
```

👉 Create **2 qubits (2 coins)**

Initial state:

---

4. Superposition

```python
qc.  0
```

👉 Apply **H gate** to qubit 0

Before:

After:

Think:
👉 Coin is spinning

---

5. Entanglement

```python
qc.   0  1
```

👉 Rule:

- If qubit 0 = 1 → flip qubit 1

Now result becomes:

[redacted]

Meaning:
👉 If first is 0 → second is 0
👉 If first is 1 → second is 1

✅ Now both qubits are **linked**

---

6. Different Bell States

```python
if state_type == "phi_minus"
    qc.  0
```

👉 Slight sign change (ignore physically for now)

```python
elif state_type == "psi_plus"
    qc.  1
```

👉 Flip second qubit

Now state becomes:

[redacted]

```python
elif state_type == "psi_minus"
    qc.  1
    qc.  0
```

👉 Flip + sign change

---

7. Return

```python
return qc
```

---

8. Loop

```python
states =  "phi_plus"  "phi_minus"  "psi_plus"  "psi_minus"
```

```python
for s in states
```

👉 Try all 4 Bell states

```python
qc = create_bell_state s
```

```python
sv = Statevector.                qc
```

👉 Show exact quantum state

---

9. Output Meaning

Example:

```
[redacted]
```

This means:

[redacted]

🔴 SUMMARY (Bell States)

Name	Meaning
phi+	00 + 11
phi−	00 − 11
psi+	01 + 10
psi−	01 − 10

🔴 PART 2 — TELEPORTATION (MAIN CONCEPT)

🎯 Goal

👉 Send state of qubit 0 → qubit 2
WITHOUT sending the qubit itself

---

1. Create qubits

```python
qr = QuantumRegister 3
```

👉 3 qubits:

- q0 → original data
- q1 → helper
- q2 → receiver

```python
cr = ClassicalRegister 2
```

👉 Stores measurement results

```python
qc = QuantumCircuit qr  cr
```

2. Create unknown state

```python
theta = np.   / 4
qc.   theta  0
```

👉 Rotate qubit 0

Now it becomes:

[redacted]

Think:
👉 Unknown data

3. Create entangled pair

```python
qc.  1
qc.   1  2
```

👉 q1 and q2 are now linked

4. Mix q0 with q1

```python
qc.   0  1
qc.  0
```

👉 This step:

- Transfers information into entanglement

5. Measure

```python
qc.        0  1    0  1
```

👉 Convert q0 and q1 → classical bits

⚠️ Important:

- After this → q0 is destroyed

---

6. Fix qubit 2

```python
qc.  2 .     cr  1
qc.  2 .     cr  2
```

👉 Based on measurement:

- Apply correction

🧠 Result

👉 q2 now has SAME state as original q0

🔴 KEY IDEA (VERY IMPORTANT)

Teleportation =

[redacted]

🔴 WHY OUTPUT LOOKS RANDOM?

[redacted]

👉 Because measurement is random
👉 But correction ensures final state is correct

🔴 FINAL INTUITION

Before:

[redacted]

After:

[redacted]

🔴 MOST COMMON CONFUSION (Fix this)

❌ "Data travels instantly"

No.

✔️ What actually happens:

1. Entanglement already exists
2. Measurement sends classical info
3. Receiver reconstructs state

🔴 If still unclear, answer this:

Tell me:

- Which step confused you most?

Options:

1. Superposition (`H gate`)
2. Entanglement (`CX`)
3. Statevector output
4. Teleportation logic

I'll drill only that part deeply (with diagrams or simplified code).

Where is phi plus given

Short answer: `phi_plus` **is not written explicitly anywhere in the code — it is the *default state* created when no extra operation is applied.**

Okay now' provide related theory for me to prepare for viva
Give basic gates info too

Below is a **viva-ready theory pack** covering Bell states, teleportation, and basic quantum gates. It's structured so you can answer both **"what" and "why"** questions concisely.

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1) Core Concepts (very likely viva questions)

Qubit

- A qubit is a 2-level quantum system:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

- Measurement collapses it to `0` with probability $|\alpha|^2$ or `1` with $|\beta|^2$.

Superposition

- A qubit can be in a linear combination of basis states.
- Created typically using the **Hadamard (H)** gate.

Entanglement

- Multi-qubit states where subsystems are not independent.
- Measurement outcomes are **correlated** beyond classical limits.
- Bell states are canonical maximally entangled states.

---

2) Bell States (must-know)

Definitions

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$|\phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

$$|\psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

How they are generated (from code)

Base circuit:

- Apply **H on qubit 0**
- Apply **CNOT (0 → 1)**

This gives:

$$|\phi^+\rangle$$

Then modify:

State	Extra Gates
φ^+	none
φ^-	Z on qubit 0
ψ^+	X on qubit 1
ψ^-	X on qubit 1 + Z on qubit 0

Key Properties

- Maximally entangled
- Measurement of one qubit determines the other
- Used in:
  - Quantum teleportation
  - Superdense coding
  - Quantum cryptography

---

3) Quantum Gates (basic but critical)

Single-Qubit Gates

1. Pauli-X (NOT gate)

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Flips state:
  - $|0\rangle \to |1\rangle$
  - $|1\rangle \to |0\rangle$

---

2. Pauli-Z

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- Changes **phase**:
  - $|0\rangle \to |0\rangle$
  - $|1\rangle \to -|1\rangle$

👉 Important: does NOT change measurement outcome directly

---

3. Hadamard (H)

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- Creates superposition:
  - $|0\rangle \to (|0\rangle + |1\rangle)/\sqrt{2}$

---

4. Rotation Gate (RY)

$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

- Used to create arbitrary states

Two-Qubit Gate

Controlled-NOT (CNOT / CX)

- If control = 1 → flip target

Truth table:

Control	Target	Output
0	0	00
0	1	01
1	0	11
1	1	10

👉 Essential for **entanglement**

---

4) Quantum Teleportation (high probability viva topic)

Goal

Transfer quantum state from qubit A → qubit C
without physically sending the qubit

Steps (map to your code)

Step 1: Prepare unknown state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

(using `RY`)

Step 2: Create Bell pair

- Entangle qubit 1 and 2

Step 3: Bell measurement

- Apply:
  - CNOT (0 → 1)
  - H on qubit 0

Step 4: Measure

- Collapse qubits 0 and 1 → classical bits

Step 5: Conditional correction

- Apply X and/or Z on qubit 2 based on result

Final Result

- Qubit 2 now contains original state
- Qubit 0 is destroyed

Important Insight (exam critical)

Teleportation uses:

$$\text{Entanglement} + \text{Classical Communication}$$

NOT faster-than-light transfer

---

5) Statevector (what you printed)

Example:

[redacted]

Means:

$$0.707|00\rangle + 0.707|11\rangle$$

Order:

[redacted]

---

6) Common Viva Questions + Answers

Q1: Why use Hadamard?

→ To create superposition

Q2: Why use CNOT?

→ To create entanglement

Q3: What is difference between X and Z?

- X flips state
- Z changes phase

Q4: Why measurement is important?

→ Converts quantum info to classical info
→ Collapses state

Q5: Why teleportation needs classical bits?

→ To send correction information

Q6: Can teleportation send matter?

→ No, only quantum state

Q7: Why Bell states are important?

→ They are maximally entangled and used as a resource

---

7) Common Mistakes (avoid in viva)

- Saying "qubit is both 0 and 1 physically"
  → say "probability amplitude superposition"
- Saying "Z gate changes value"
  → it changes **phase only**
- Saying "teleportation is instant transfer"
  → requires classical communication

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8) If you want higher-level prep

I can push you to next level:

- Dirac notation derivation (step-by-step teleportation math)
- Matrix multiplication walkthrough
- Typical viva trick questions (professor-level)
- Short 2-mark / 5-mark answers

Tell me your exam format.